

Customer Shopping Behavior Analysis

3,900 transactions · 18 features · Python · SQL · Power BI



Dataset at a Glance

3,900

Transactions

18

Features

37

Missing Values

Review Rating column



Key Features

- Demographics: Age, Gender, Location, Subscription
- Purchases: Item, Category, Amount, Season
- Behavior: Discount, Promo Code, Frequency, Rating



Data Preparation in Python



Clean & Load

Imported via pandas, checked structure with `info()` and `describe()`



Impute & Standardize

Median imputation for Review Rating; renamed columns to snake_case



Engineer Features

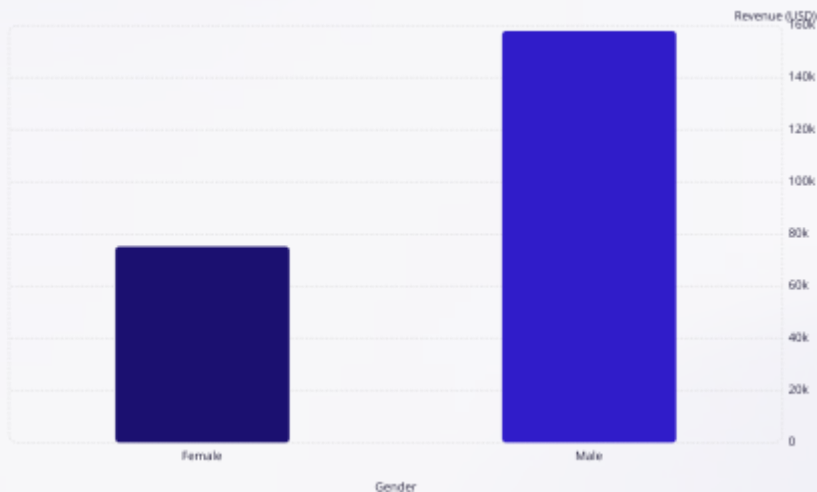
Created `age_group` and `purchase_frequency_days`; dropped redundant `promo_code_used`



Load to PostgreSQL

Cleaned DataFrame loaded into database for SQL analysis

Revenue by Gender



Key Insight

Male customers generated **2.1× more revenue** than female customers — \$157,890 vs \$75,191.



Male shoppers represent 68% of customers (2,652 of 3,900).

Top Products by Rating & Discount Dependency

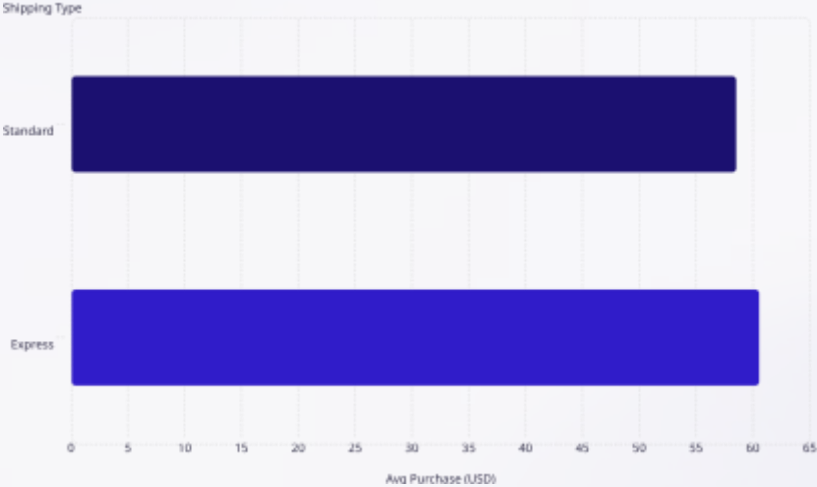
Top 5 by Average Rating

Product	Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Highest Discount Rates

Product	Discount %
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Shipping & Subscription Insights

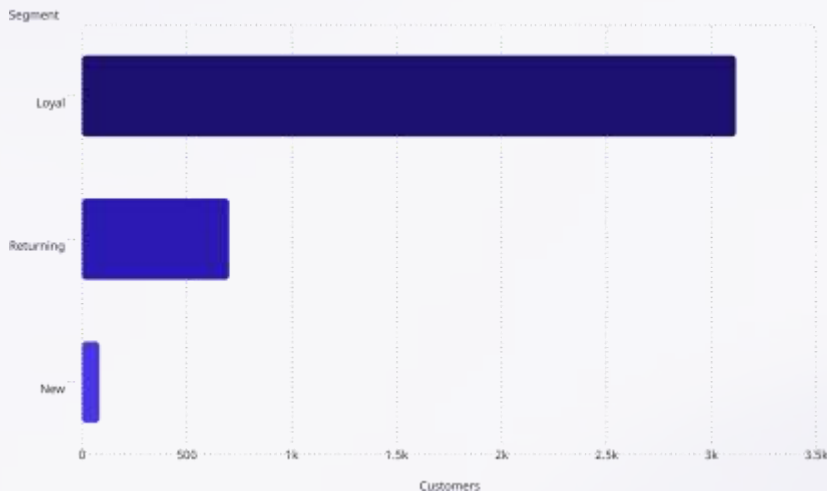


Subscribers vs Non-Subscribers

Status	Customers	Avg Spend	Total Revenue
Yes	1,053	\$59.49	\$62,645
No	2,847	\$59.87	\$170,436

Non-subscribers drive **73% of total revenue**.

Customer Segmentation



Segment Breakdown

Loyal — 80%

3,116 customers

Returning — 18%

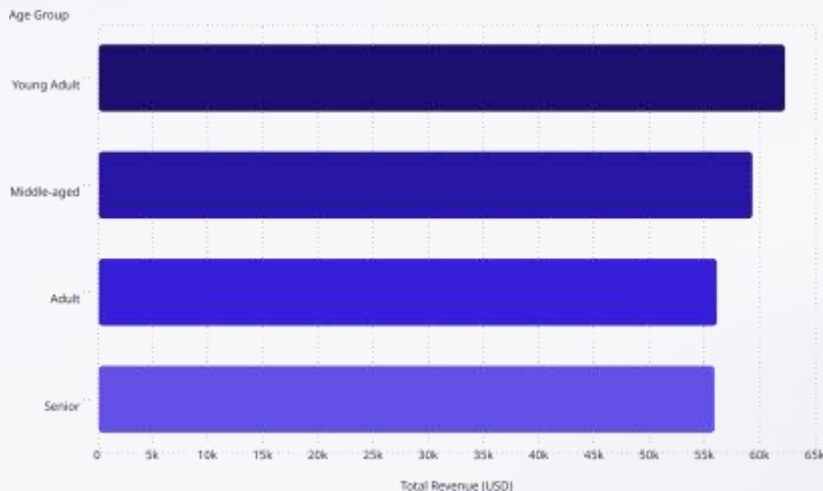
701 customers

New — 2%

83 customers

Repeat buyers with 5+ purchases: **958 subscribers** vs **2,518 non-subscribers**.

Revenue by Age Group



Age Range

Customers aged **18–70**, mean age **44**.

Young Adults lead revenue at **\$62,143**, with relatively even distribution across all groups.

📄 Average purchase amount: **\$59.76** · Average rating: **3.75**

Power BI Dashboard

KPIs

3.9K customers · \$59.76 avg
purchase · 3.75 avg rating

Filters

Subscription · Gender ·
Category · Shipping Type

Visuals

Revenue by Category & Age Group · Subscription Donut · Sales Charts



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers toward "Loyal" segment



Review Discount Policy

Balance sales boosts with margin control



Targeted Marketing

Focus on high-revenue age groups & express-shipping users



Product Positioning

Highlight top-rated & best-selling products in campaigns